

our state estimator might be overly susceptible to modeling errors.⁴ When designing a model for a Kalman filter, we need to balance our confidence in our model (low noise resulting in close model tracking; i.e., low bandwidth) with a healthy self-doubt (high noise resulting in filter responsiveness; i.e., high bandwidth).

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Examination of the filter equations shows why adding fictitious process noise compensates for modeling errors. Recall the Kalman filter equations from Equation (5.19), some of which we repeat here:

$$\begin{aligned} P_k^- &= F_{k-1}P_{k-1}^+F_{k-1}^T + Q_{k-1} \\ K_k &= P_k^-H_k^T(H_kP_k^-H_k^T + R_k)^{-1} \\ \hat{x}_k^+ &= \hat{x}_k^- + K_k(y_k - H_k\hat{x}_k^-) \end{aligned} \quad (5.63)$$

If Q_k is small then the covariance may not increase very much between time samples. In Example 5.3 we had $F_k = 1$, so $P_k^- = P_{k-1}^+$ when $Q_k = 0$. But the covariance will decrease from P_k^- down to P_k^+ every time a measurement is obtained due to the measurement-update equation for the covariance. Eventually P_k^- will converge to zero. This can be seen by looking at Equation (5.26), which shows the one-step equation for P_k^- :

$$P_{k+1}^- = F_kP_k^-F_k^T - F_kK_kH_kP_k^-F_k^T + Q_k \quad (5.64)$$

If $Q_k = 0$ then this equation has a steady solution of zero. A zero value for P_k^- will result in $K_k = 0$, as seen from Equation (5.63). A zero value for K_k means that the measurement-update equation (5.63) for $\hat{x}$ will not take any account of the measurement – that is, the measurement y_k will be completely ignored in the computation of $\hat{x}^+$. This is because the measurement noise covariance R_k (assuming it is greater than zero) will be infinitely times larger than the process noise $Q_k = 0$. The filter will become sluggish in the sense that it will not respond to measurements.

On the other hand, if Q_k is larger, then the covariance will always increase between time samples – that is, P_k^- will always be larger than P_{k-1}^+ . When P_k^- converges, it will converge to a larger value. This will make K_k converge to a larger value. A larger K_k means that the measurement update for $\hat{x}$ in Equation (5.63) will include a larger emphasis on the measurement – that is, the filter will pay more attention to the measurements.

5.6 SUMMARY

In this chapter, we have presented the essence of the discrete-time Kalman filter. Over the past few decades, this estimation algorithm has found applications in virtually every area of engineering. We have seen that the Kalman filter equations can be written in several different ways, each of which may appear quite different than the others, although they are all mathematically equivalent. We have seen that

⁴Noise, like most things in life, is beneficial in moderate amounts. We also see this in human psychological responses to noise. Too much noise will drive humans insane, but too little noise might also result in a loss of sanity. Noise is especially beneficial for controls engineers, who would not only lose their sanity but would also lose their research funding if not for noise [Bar01, p. 179].

the Kalman filter is optimal even when the noise is not Gaussian. The Kalman filter is the optimal estimator when the noise is Gaussian, and it is the optimal *linear* estimator when the noise is not Gaussian. We have seen that the Kalman filter may not perform well if the underlying assumptions do not hold, and we briefly mentioned some ways to compensate for violated assumptions. The later chapters of this book will expand and generalize the results presented in this chapter.

PROBLEMS

Written exercises

5.1 A radioactive mass has a half-life of τ seconds. At each time step the number of emitted particles x is half of what it was one time step ago, but there is some error w_k (zero-mean with variance Q) in the number of emitted particles due to background radiation. At each time step, the number of emitted particles is counted. The instrument used to count the number of emitted particles has a random error at time k of v_k , which is zero-mean with a variance of R . Assume that w_k and v_k are uncorrelated.

- Write the linear system equations for this system.
- Suppose we want to use a Kalman filter to find the optimal estimate of the number of emitted particles at each time step. Write the one-step *a posteriori* Kalman filter equations for this system.
- Find the steady-state *a posteriori* estimation-error variance for the Kalman filter.
- What is the steady-state Kalman gain when $Q = R$? What is the steady-state Kalman gain when $Q = 2R$? Give an intuitive explanation for why the steady-state gain changes the way it does when the ratio of Q to R changes.

5.2 This problem illustrates the robustness that is achieved by the use of the Joseph form of the covariance measurement update equation. Suppose you have a discrete-time Kalman filter for a scalar system.

- Find $\partial P_k^+ / \partial K_k$ for the third form of the covariance measurement update in Equation (5.19).
- Find $\partial P_k^+ / \partial K_k$ for the Joseph form (the first form) of the covariance measurement update in Equation (5.19). After you get your answer, substitute for K_k from the Kalman gain expression.
- Use the above results to explain why the Joseph form of the covariance measurement-update equation is stable and robust.

5.3 Prove that $E[\hat{x}_k^+ (\hat{x}_k^+)^T] = 0$. Hint: Since $\hat{x}_0^+ = E[x_0]$ is a constant and $\tilde{x}_0^+ = x_0 - \hat{x}_0^+$ is zero-mean, we know that $E[\hat{x}_0^+ (\tilde{x}_0^+)^T] = 0$. Given this information, prove that $E[\hat{x}_1^+ (\tilde{x}_1^+)^T] = 0$. From this point, use induction to complete the proof.

5.4 Suppose that you have a fish tank with x_p piranhas and x_g guppies [Bay99]. Once per week, you put guppy food into the tank (which the piranhas do not eat). Each week the piranhas eat some of the guppies. The birth rate of the piranhas is proportional to the guppy population, and the death rate of the piranhas is

proportional to their own population (due to overcrowding). Therefore $x_p(k+1) = x_p(k) + k_1 x_g(k) - k_2 x_p(k) + w_p(k)$, where k_1 and k_2 are proportionality constants and $w_p(k)$ is white noise with a variance of one that accounts for mismodeling. The birth rate of the guppies is proportional to the food supply u , and the death rate of the guppies is proportional to the piranha population. Therefore, $x_g(k+1) = x_g(k) + u(k) - k_3 x_p(k) + w_g(k)$, where k_3 is a proportionality constant and $w_g(k)$ is white noise with a variance of one that accounts for mismodeling. The step size for this model is one week. Every week, you count the piranhas and guppies. You can count the piranhas accurately because they are so large, but your guppy count has zero-mean noise with a variance of one. Assume that $k_1 = 1$ and $k_2 = k_3 = 1/2$.

- Generate a linear state-space model for this system.
- Suppose that at the initial time you have a perfect count for x_p and x_g . Using a Kalman filter to estimate the guppy population, what is the variance of your guppy population estimate after one week? What is the variance after two weeks?
- What is the ratio of the piranha population to the guppy population when they reach steady state? Assume that the process noise is zero for this part of the problem.

5.5 The measured output of a simple moving average process is $y_k = z_k + z_{k-1}$, where $\{z_j\}$ is zero-mean white noise with a variance of one.

- Generate a state-space description for this system with the first element of x_k equal to z_{k-1} and second element equal to z_k .
- Suppose that the initial estimation-error covariance is equal to the identity matrix. Show that the *a posteriori* estimation-error covariance is given by

$$P_k^+ = \frac{1}{k+1} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

- Find $E[||x_k - \hat{x}_k^+||_2^2]$ as a function of k .

5.6 In this problem, we use the auxiliary variable $S_k = H_k P_k^- H_k^T + R_k$. Note that

$$\begin{bmatrix} I & 0 \\ -P_k^- H_k^T S_k^{-1} & I \end{bmatrix} \begin{bmatrix} S_k & H_k P_k^- \\ P_k^- H_k^T & P_k^- \end{bmatrix} = \begin{bmatrix} S_k & H_k P_k^- \\ 0 & P_k^+ \end{bmatrix}$$

Use the product rule for determinants to show that

$$|P_k^+| = \frac{|P_k^-| |R_k|}{|S_k|}$$

5.7 In Section 4.1, we saw that Σ_k , the covariance of the state of a discrete-time system, is given as $\Sigma_{k+1} = F_k \Sigma_k F_k^T + Q_k$. Use this along with the one-step expression for the *a priori* estimation-error covariance of the Kalman filter to show that $\Sigma_k - P_k^- \geq 0$ for all k . Give an intuitive explanation for this expression [And79].

5.8 Consider the system of Problem 5.1.

- Use the method of Section 5.4 to find a closed-form solution for P_k^- , assuming that $Q = 1$, $R = 5$, and $P_0 = 0$.
- Use your result from above to find the steady-state value of P_k^- .

5.9 Suppose that a Kalman filter is designed for the system

$$\begin{aligned}x_{k+1} &= x_k \\y_k &= x_k + v_k \\v_k &\sim (0, R)\end{aligned}$$

- a) Suppose that $E(x_0^2) = 1$. Design a Kalman filter for the system and find a closed-form expression for P_k^- . What is the limit of P_k^- as $k \rightarrow \infty$?
- b) Now suppose that the true process equation is actually $x_{k+1} = x_k + w_k$, where $w_k \sim (0, Q)$. Find a difference equation for the variance of the *a priori* estimation error if the Kalman filter that you designed in part (a) is used to estimate the state. What is the limit of the estimation-error variance as $k \rightarrow \infty$?

5.10 Suppose that a Kalman filter is designed for a discrete LTI system with an assumed measurement noise covariance of R , but the actual measurement noise covariance is $(R + \Delta R)$. The output of the Kalman filter will indicate that the *a priori* estimation-error covariance is P_k^- , but the actual *a priori* estimation-error covariance will be Σ_k^- . Find a difference equation for $\Delta_k = (\Sigma_k^- - P_k^-)$. Will Δ_k always be positive definite?

Computer exercises

5.11 Let p_k denote the wombat population at time k , and f_k denote the size of the wombat's food supply at time k . From one time step to the next, half of the existing wombat population dies, but the number of new wombats is added to the population is equal to twice the food supply. The food supply is constant except for zero-mean random fluctuations with a variance of 10. At each time step the wombat population is counted with an error that has zero mean and a variance of 10. The initial state is

$$\begin{aligned}p_0 &= 650 \\f_0 &= 250\end{aligned}$$

The initial state estimate and uncertainty is

$$\begin{aligned}\hat{p}_0 &= 600 \\E[(\hat{p}_0 - p_0)^2] &= 500 \\\hat{f}_0 &= 200 \\E[(\hat{f}_0 - f_0)^2] &= 200\end{aligned}$$

Design a Kalman filter to estimate the population and food supply.

- a) Simulate the system and the Kalman filter for 10 time steps. Hand in the following.
 - Source code listing.
 - A plot showing the true population and the estimated population as a function of time.

- A plot showing the true food supply and the estimated food supply as a function of time.
 - A plot showing the standard deviation of the population and food supply estimation error as a function of time.
 - A plot showing the elements of the Kalman gain matrix as a function of time.
- b) Compare the standard deviation of the estimation error of your simulation with the steady-state theoretical standard deviation based on P_k^+ . Why is there such a discrepancy?
- c) Run the simulation again for 1000 time steps and compare the experimental estimation error standard deviation with the theoretical standard deviation.

5.12 Consider the RLC circuit described in Problem 1.18 with $R = 3$, $L = 1$, and $C = 0.5$. The input voltage is zero-mean, unity variance white noise. Suppose that the capacitor voltage is measured at 10 Hz with zero-mean, unity variance white noise. Design a Kalman filter to estimate the inductor current, with an initial covariance $P_0^+ = 0$. Generate a plot showing the *a priori* and *a posteriori* variances of the inductor current estimate for 20 time steps. Based on the plot, what is the steady-state value of P_k^- ? Use the development of Section 5.4.1 to approximate the steady-state value of P_k^- using 1, 2, 3, and 4 successive squares of the Ψ matrix.